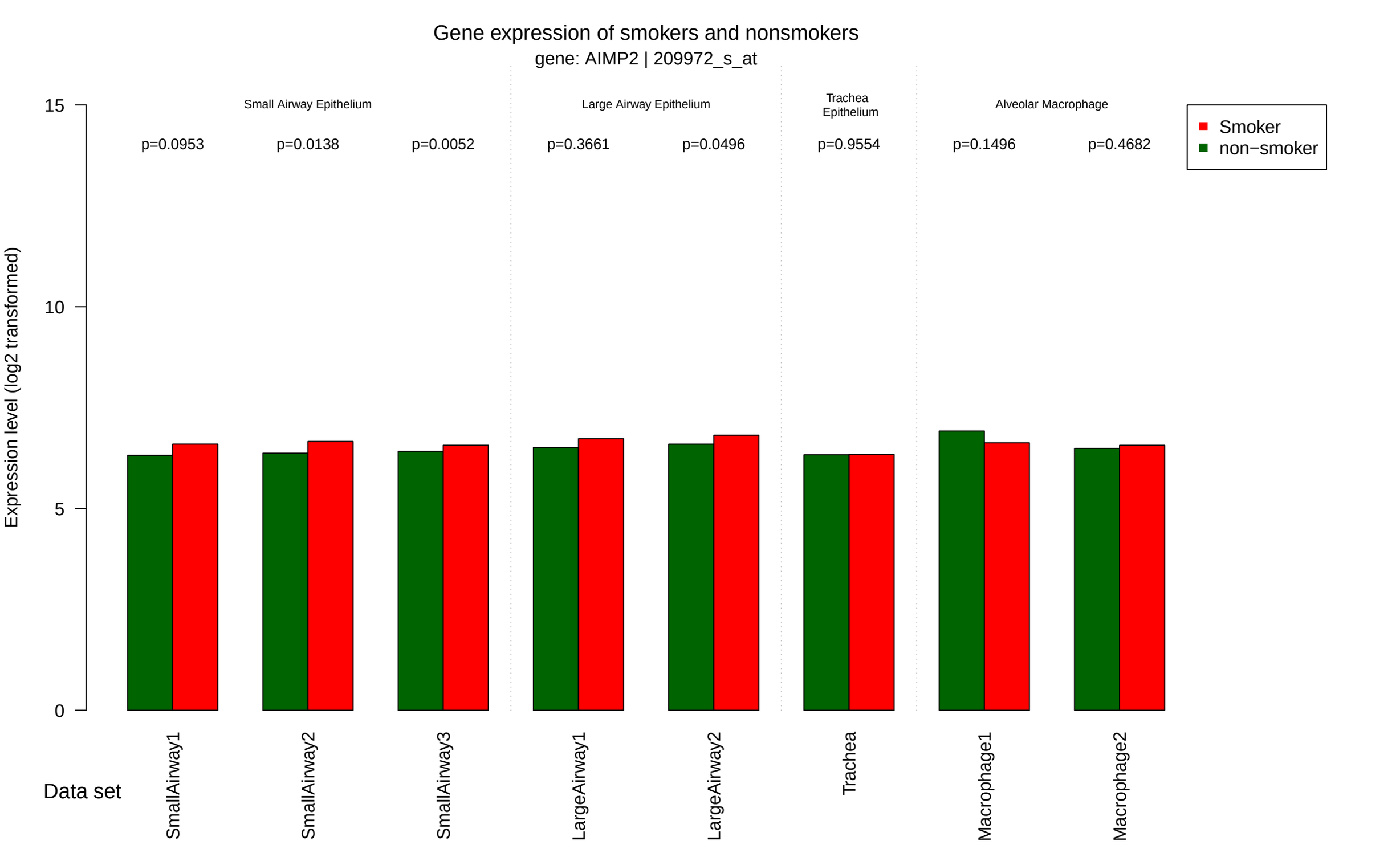
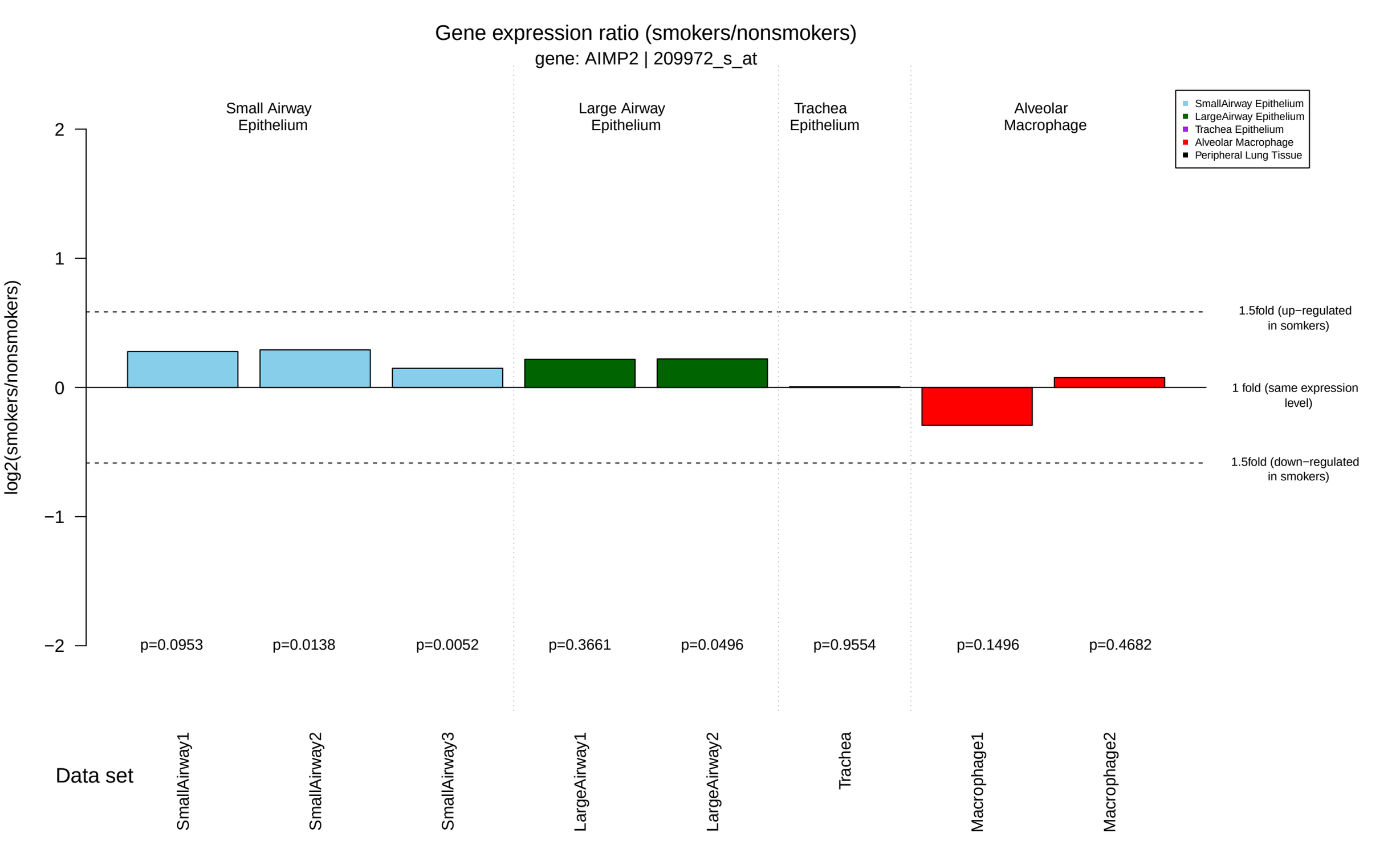
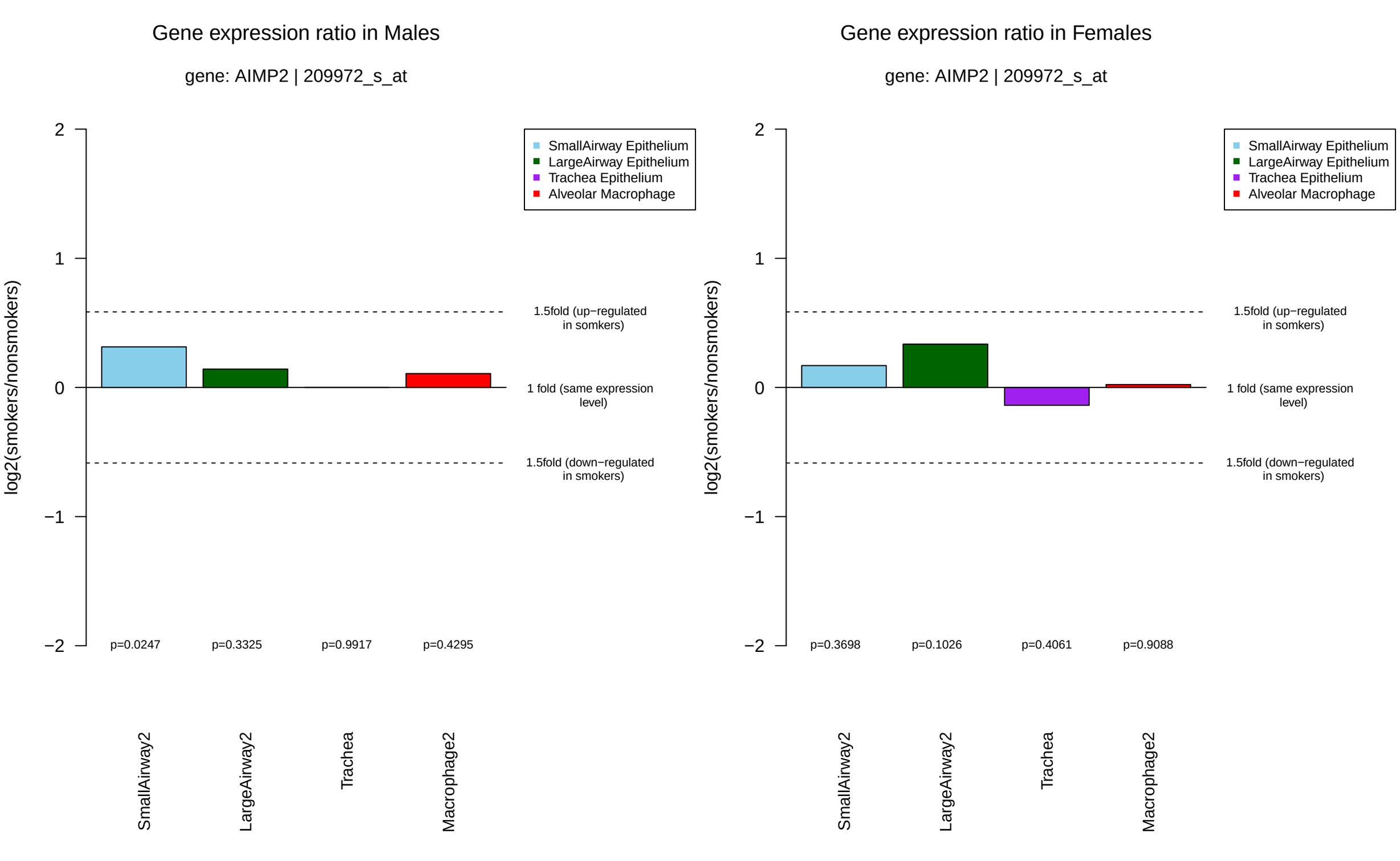
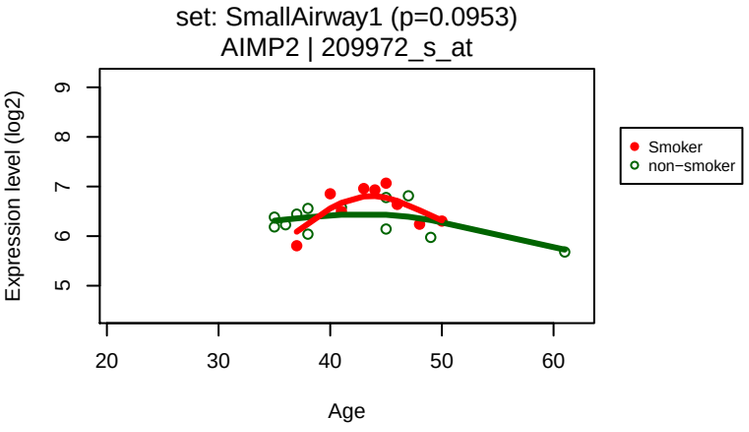
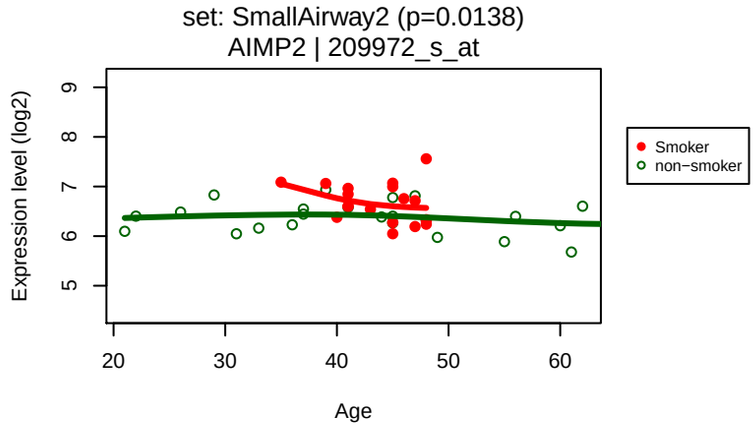
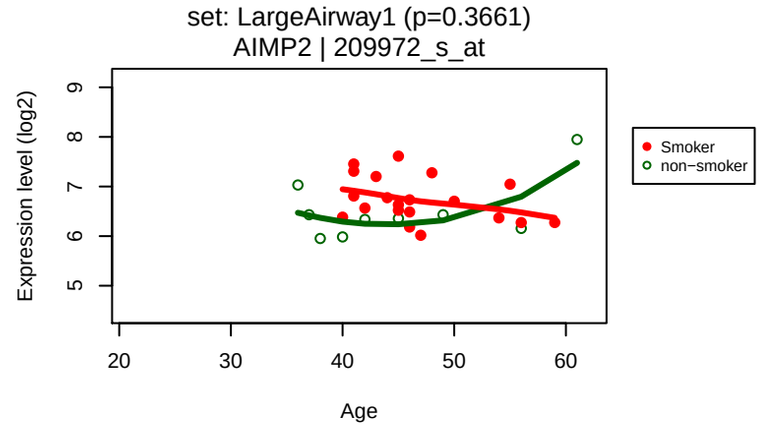

Gene expression of smokers and nonsmokers
gene: AIMP2 | 209972_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0953
p=0.0138
p=0.0052
p=0.3661
p=0.0496
p=0.9554
p=0.1496
p=0.4682
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

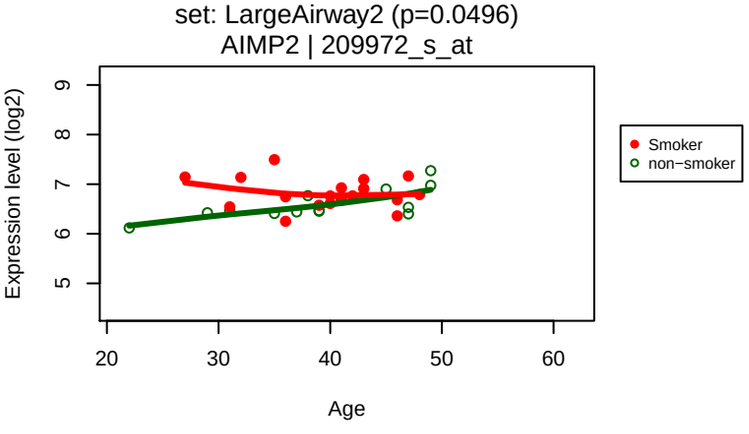

Gene expression ratio (smokers/nonsmokers)
gene: AIMP2 | 209972_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0953
p=0.0138
p=0.0052
p=0.3661
p=0.0496
p=0.9554
p=0.1496
p=0.4682
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

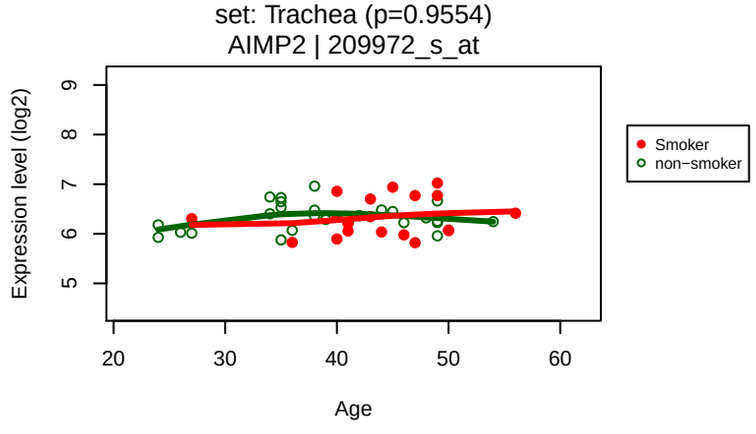

Gene expression ratio in Males
gene: AIMP2 | 209972_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0247
p=0.3325
p=0.9917
p=0.4295
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: AIMP2 | 209972_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3698
p=0.1026
p=0.4061
p=0.9088
SmallAirway2
LargeAirway2
Trachea
Macrophage2

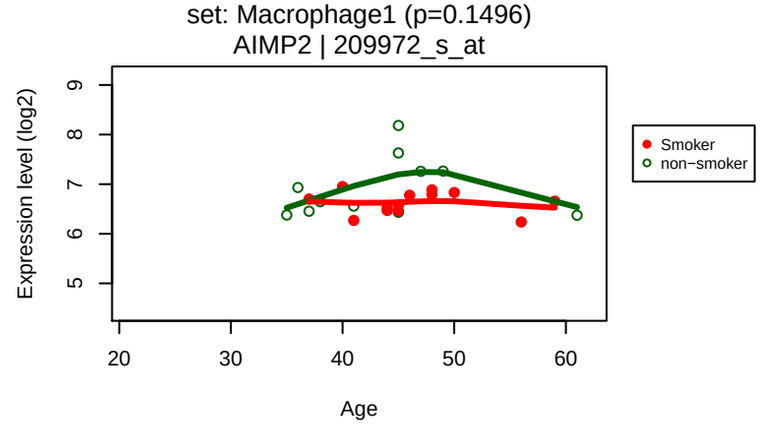

set: SmallAirway1 (p=0.0953)
AIMP2 | 209972_s_at
Expression level (log2)
Age
Smoker
non−smoker

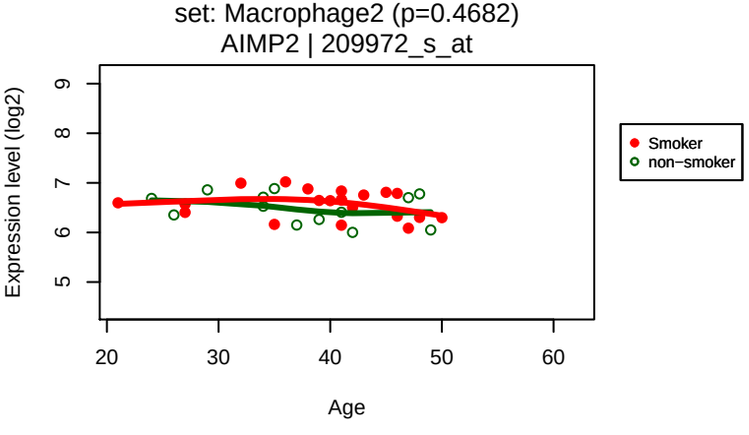

set: SmallAirway2 (p=0.0138)
AIMP2 | 209972_s_at
Expression level (log2)
Age
Smoker
non−smoker

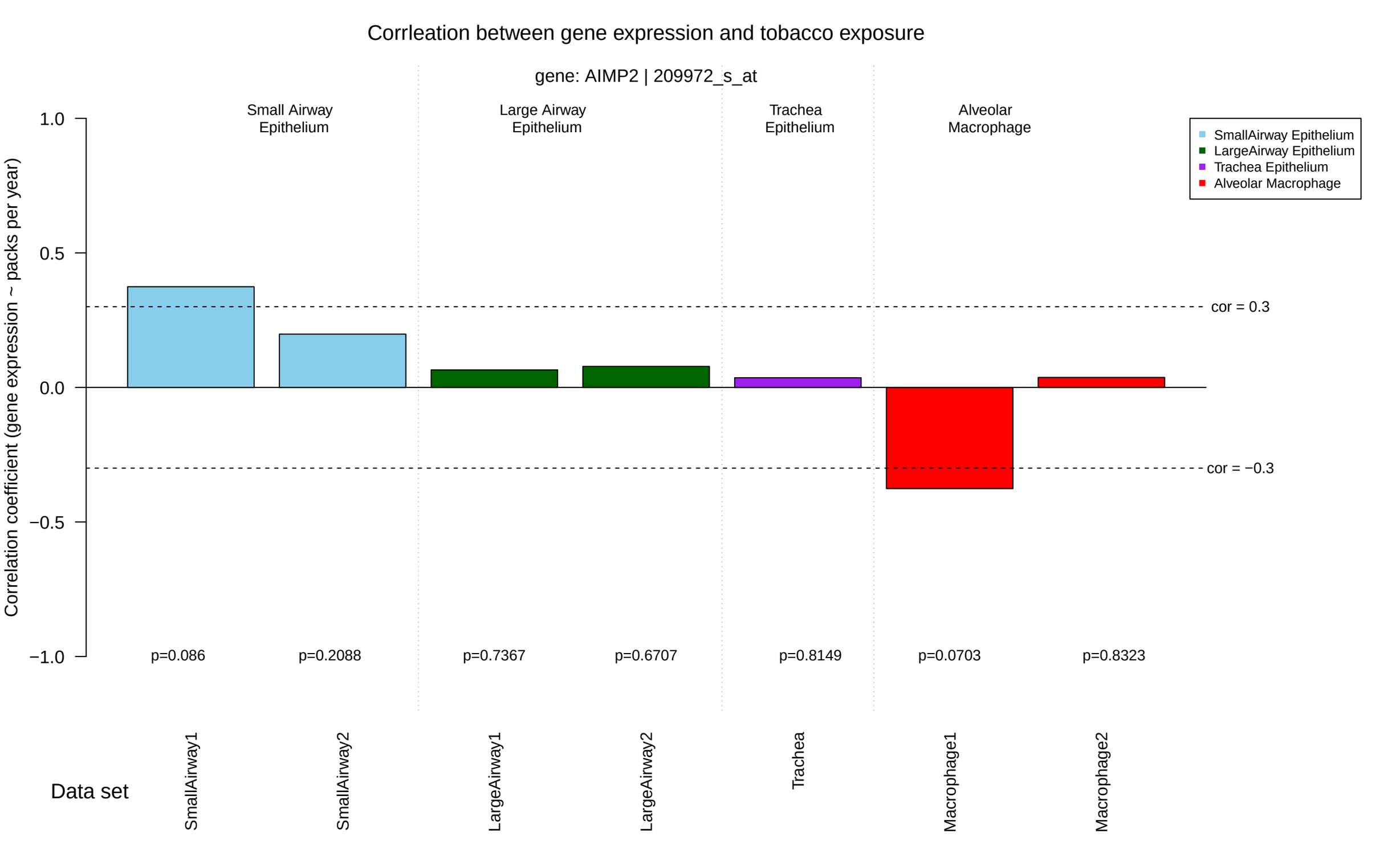

set: LargeAirway1 (p=0.3661)
AIMP2 | 209972_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0496)
AIMP2 | 209972_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9554)
AIMP2 | 209972_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1496)
AIMP2 | 209972_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4682)
AIMP2 | 209972_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: AIMP2 | 209972_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.086
p=0.2088
p=0.7367
p=0.6707
p=0.8149
p=0.0703
p=0.8323
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2